

Dilations and Similar Figures

Answer Key

Grade 8 / Pre-Algebra

Quick Answers

- | | |
|--|---|
| 1. the x-coordinate of $A' = 6$, the y-coordinate of $A' = 9$ | 7. the slope of $DE = 2$, the slope of $D'E' = 2$, the scale factor $k = 3$ |
| 2. the length of $PQ = 5$, the length of $P'Q' = 10$ | 8. the factor along the horizontal side $= 2$, the factor along the vertical side $= 3$, — |
| 3. the scale factor $k = \frac{5}{2}$, the longer side of the enlargement $= 22.5$ cm | 9. the real span $= 128$ m, the real tower height $= 18$ m |
| 4. the slope of $AB = 2$, the slope of $A'B' = 2$, — | 10. the length of $KL = 10$, the perimeter of $JKL = 24$, the perimeter of the image $= 36$, — |
| 5. the length of the image side $A'B' = 8$, — | |
| 6. 30 cm | |
-

What is verified

6 of 10 problems fully machine-checked against SymPy.

— marks an answer only you can judge — problems 4, 5, 8, 10. The worked solution states what a correct response must contain.

Curriculum

Level: Grade 8 / Pre-Algebra

8.G.A – problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10

Difficulty 1–5 of 5 across 24 tagged checks.

- 1.** $A(2, 3)$ is dilated about the origin with $k = 3$. Find A' .

Solution: A dilation about the origin multiplies both coordinates by k , so $x' = 3 \cdot 2 = 6$ and $y' = 3 \cdot 3 = 9$. $A'(6, 9)$

- 2.** $P(1, 2)$, $Q(4, 6)$; images $P'(2, 4)$, $Q'(8, 12)$ under $k = 2$. Find PQ and $P'Q'$.

Solution: Use the distance formula on each pair.

$$PQ = \sqrt{(4-1)^2 + (6-2)^2} = \sqrt{9+16} = \sqrt{25}$$

$$P'Q' = \sqrt{(8-2)^2 + (12-4)^2} = \sqrt{36+64} = \sqrt{100}$$

The image is exactly twice as long, which is what the scale factor 2 promises.

$PQ = 5$	$P'Q' = 10$
----------	-------------

3. Print 6 cm by 9 cm; similar enlargement is 15 cm on the short side. Find k and the long side.

Solution: The scale factor is the image length divided by the matching original length, taken on the pair we know: $k = \frac{15}{6}$, which reduces to $\frac{5}{2}$ (that is 2.5). Apply the same factor to the long side: $9 \cdot \frac{5}{2} = \frac{45}{2}$.

$$k = \frac{5}{2} \quad \text{long side} = 22.5 \text{ cm}$$

4. $A(1, 2)$, $B(4, 8)$; images $A'(0.5, 1)$, $B'(2, 4)$ under $k = \frac{1}{2}$. Compare the slopes.

Solution: Slope is rise over run.

$$\text{slope } AB = \frac{8 - 2}{4 - 1} = \frac{6}{3}, \quad \text{slope } A'B' = \frac{4 - 1}{2 - 0.5} = \frac{3}{1.5}$$

Both reduce to the same number.

$$\text{slope } AB = 2 \quad \text{slope } A'B' = 2$$

Reason (instructor-judged). A correct response says that equal slopes on two different segments mean the segments are parallel, and that a dilation about the origin always produces a parallel image of a segment that does not pass through the centre. Accept any wording that gets from “same slope” to “parallel”; two numbers with no conclusion is not a reason.

5. Dilate $A(-3, -1)$, $B(1, -1)$, $C(1, 2)$ by $k = 2$ about the origin; draw the image and find $A'B'$.

Drawing (instructor-judged). Doubling both coordinates gives $A'(-6, -2)$, $B'(2, -2)$ and $C'(2, 4)$. Full credit for a triangle drawn at exactly those three points with the vertices labelled and the sides joined. A correctly shaped triangle in the wrong place, or image vertices that are not exactly twice the originals, loses credit.

Solution: $A'B'$ is horizontal, so its length is the difference of the x -coordinates: $|2 - (-6)| = 8$.

$$A'B' = 8$$

6. A photograph 8 cm wide and 12 cm tall is enlarged to 20 cm wide. How tall is the enlargement?

Solution: Similar figures scale by one factor, so the height scales the same way the width did: $k = \frac{20}{8} = 2.5$, and the new height is $12 \cdot 2.5$.

$$30 \text{ cm}$$

7. $D(2, 1)$, $E(4, 5)$; images $D'(6, 3)$, $E'(12, 15)$. Find both slopes and the scale factor.

Solution:

$$\text{slope } DE = \frac{5 - 1}{4 - 2} = \frac{4}{2}, \quad \text{slope } D'E' = \frac{15 - 3}{12 - 6} = \frac{12}{6}$$

Both equal 2, so the segments are parallel. The scale factor comes from any pair of matching coordinates: $k = \frac{6}{2}$.

slope $DE = 2$	slope $D'E' = 2$	$k = 3$
----------------	------------------	---------

8. Is the triangle $(0, 0)$, $(8, 0)$, $(0, 9)$ a dilation of RST with $R(0, 0)$, $S(4, 0)$, $T(0, 3)$?

Solution: Compare each side with its match. Along RS : $\frac{8}{4} = 2$. Along RT : $\frac{9}{3} = 3$.

factor along $RS = 2$	factor along $RT = 3$
-----------------------	-----------------------

Judgement (instructor-judged). A correct response says the student is wrong: a dilation uses ONE scale factor in every direction, and here the horizontal is multiplied by 2 while the vertical is multiplied by 3. That is a stretch, and the second triangle is not similar to RST . Accept any wording that contrasts the two factors; “no” on its own earns no credit.

9. A model bridge scales up by 40: span 3.2 m, tower 0.45 m. Find the real lengths.

Solution: Every real length is 40 times the model length, so multiply each measurement by 40: $3.2 \cdot 40 = 128$ and $0.45 \cdot 40 = 18$.

span = 128 m	tower = 18 m
--------------	--------------

10. $J(0, 0)$, $K(6, 0)$, $L(0, 8)$ dilated by $k = 1.5$. Find KL , the perimeter of JKL , and the perimeter of the image.

Solution: $JK = 6$ and $JL = 8$ are along the axes. For KL ,

$$KL = \sqrt{(0 - 6)^2 + (8 - 0)^2} = \sqrt{36 + 64} = \sqrt{100} = 10$$

so the perimeter of JKL is $6 + 8 + 10 = 24$. Every image side is 1.5 times its original, so the image perimeter is $24 \cdot 1.5 = 36$.

$KL = 10$	perimeter $JKL = 24$	perimeter $J'K'L' = 36$
-----------	----------------------	-------------------------

Reason (instructor-judged). A correct response says that because every side is multiplied by 1.5, the sum of the sides is multiplied by 1.5 too, so the image perimeter is $24 \cdot 1.5 = 36$ without finding a single image coordinate. Accept any wording that applies the factor to the whole perimeter; recomputing the coordinates without saying why the shortcut works earns no credit.